

Soft-Settle Micro-Credit Line Build Checklist

Blockchain Layer-

Nitrolite State Channel Contract (Yellow Network Integration):

- Develop bidirectional state channels for off-chain micro-authorizations.
- Implement collateral lock: Consumer deposits USDC (e.g., 50 USDC), emit event, generate session ID.
- Handle initial state: Balance = deposit, Counter/Nonce = 0.
- Support session parameters: Max duration (e.g., 1 hour), micro-payment amount (0.001 USDC), unit definition (e.g., per URL).
- Implement mutual closure: Both sign final state (e.g., "Total transferred: 14.50 USDC"), release funds.
- Add unilateral closure: Challenge period (e.g., 10 minutes) for unresponsive parties, on-chain enforcement.
- Use EIP-712 for typed data signing of state updates (e.g., "Shift 0.001 USDC; New balance: X; Nonce: Y").
- Optimize gas: Target <200k gas per open/close.

Reputation Manager Contract:

- Automate ENS text record updates: Keys like vnd.credit.score (0-1000), vnd.deals.total (USDC volume), vnd.disputes (count).
- Implement score calculation: e.g., New Score = (Previous * 0.9) + (100 if successful, -200 if disputed).
- Hook into settlement events for post-session updates.
- Ensure access to settlement data for calculations.

Integration Hooks:

- Add callbacks for Circle APIs (payouts if deficit > deposit).
- Use Solidity 0.8.x; develop/test with Hardhat.

Deployment and Testing:

- Deploy to testnet (e.g., Polygon Mumbai) first.
- Unit tests: Cover locking, signing, closures, disputes.

- Audit: Plan for external audit (e.g., PeckShield) for reentrancy, overflows.

Off-Chain Layer (Microservices and Messaging)

P2P Messaging System:

- Build using Node.js/Go: Handle real-time communication (e.g., libp2p or WebSockets) for work delivery (data hash/payload) and micro-authorizations.
- Ensure loop speed: <100ms for work → sign → validate.
- Track cumulative transfers off-chain for both agents.
- Fallback: Relay via website WebSocket if direct P2P fails.

Signature Validation:

- Verify EIP-712 signatures against channel state; use nonces to prevent replays.
- Handle errors: Close session early on invalid signatures, trigger alerts.

Session Orchestration:

- Manage work delivery loop: Provider sends unit, consumer signs update, provider validates/continues.
- Monitor for anomalies (e.g., timeouts) and log to IPFS for audits.

Scalability Features:

- Horizontal scaling: Design for multiple nodes to handle high-frequency interactions.

Backend Layer (API Server)

API Endpoints (Express.js with GraphQL/Apollo):

/register-agent: Accept ENS domain, description, API endpoint; register in DB, initialize ENS records (e.g., initial score 500).

/init-session: Handle credit check (query ENS via resolver), collateral lock, channel open.

/query-reputation: Fetch ENS keys (credit.score, deals.total, disputes); apply logic (e.g., if >700, reduce collateral).

/monitor-session: Real-time queries for progress, balances, logs.

/close-session: Submit final state, trigger payouts if deficit, update reputation.

Additional: Dispute initiation (hook to external arbitration), export reports.

Agent Autonomy Support:

Enable self-authentication via signed messages.

Database Setup: MongoDB/Postgres for agent registry, session data.

Security: Rate limiting, multi-sig for critical actions.

Testing: Unit/integration tests with Mocha/Chai; simulate 100 sessions using Ganache.

Frontend Layer (Website/UI)

Core Pages (React.js + Next.js):

- Landing Page: Clean design with "Register Agent" CTA, vision/problem overview.
- Dashboard: Agent list, editable profiles, real-time status badges (e.g., "Active", "Low Credit"). Inspired by ENS domain management.
- Agent Registration: Form for ENS domain, description, callback API; wallet connect via Web3Modal/Ethers.js.
- Session Monitor: Real-time feed (like Twitter timeline) with expandable cards (details: collateral, participants, trends charts). Live page with progress bars (like Stripe), alerts for failures.
- Analytics Tab: Graphs for settled USDC, dispute rates (like Dune); exportable audit reports.

Features:

- Wallet integration: Connect Ethereum wallet, sign transactions (e.g., collateral lock).
- Real-time updates: Use GraphQL subscriptions for session events, on-chain indexing via The Graph.
- UI Polish: Tailwind CSS for responsiveness, Framer Motion animations, Chart.js for visuals, dark/light auto-detect.
- Error Handling: Modals for confirmations (e.g., Uniswap-style success popups), notifications for failures.

API Docs: Swagger UI for programmatic agent access.

Accessibility: Mobile-responsive, internationalization support.

Testing: End-to-end with Cypress.

Integrations

ENS Integration:

Resolver for name-to-address; read/write text records (e.g., credit keys).

Use ENS.js library.

Yellow Network (Nitrolite):

SDK for state channels; fork examples for custom contracts.

Circle/Arc:

API for USDC payouts, wallet balance queries, credit limit management (auto-adjust, e.g., 2x balance).

Store API keys in env vars; trigger payouts on deficit.

Other Tools:

Alchemy/Infura for RPC.

The Graph for event indexing.

IPFS for logs.

Testing: Load testing with Artillery for scalability; beta deploy to testnet.